Project-II

Instructions

1. MATLAB is the preferred programming platform in this course.
2. Do not use any in-built or intrinsic functions, as you are expected to write your computer program. However, for solving the system of algebraic equations, you may use the in-built linear solvers in MATLAB (such as `mldivide`, '\' (The backslash operator), `linsolve`, `inv`, etc.).
3. Discussion among students is permitted.
4. You have to submit your codes as a single 'zip' file using the link/form that will be provided before the submission deadline.
5. Along with this, you must also submit a report. Your report must consist of:
   (a) Problem statement
   (b) Mesh details and approach for discretization
   (c) Derivation and presentation of the final form of the discretized equations
   (d) Solution methodology
   (e) Results and discussion, including the plots.
   (f) Concluding remarks

   Note that an analysis and discussion of results is required.
6. Parts (a) - (d) and (f) of the report must be hand-written. For Part (e), the students may simply take print out of the graphs and append them with the report. Only hard copies of the report has to be submitted to the TAs. The handwriting must be clean and easily readable.
7. Usage of any artificial intelligence (AI) tools to write project reports or develop code modules is a violation of the course policy. Strict action will be taken against those who are found to use AI for writing codes.
8. Copying codes from others is also a strict violation of the course policy.

Project Statement

Consider the heat treatment of a flat thin metallic plate (see figure) using a localized heat source such as laser. In most cases, the intense heat (loosely speaking) from the laser will melt the plate. But, for simplicity, let us assume that the plate does not melt, and rather it reaches a steady state, perhaps because either the heat source (i.e., the laser) is weak, or because there is sufficient heat transfer between the edges of the plate and

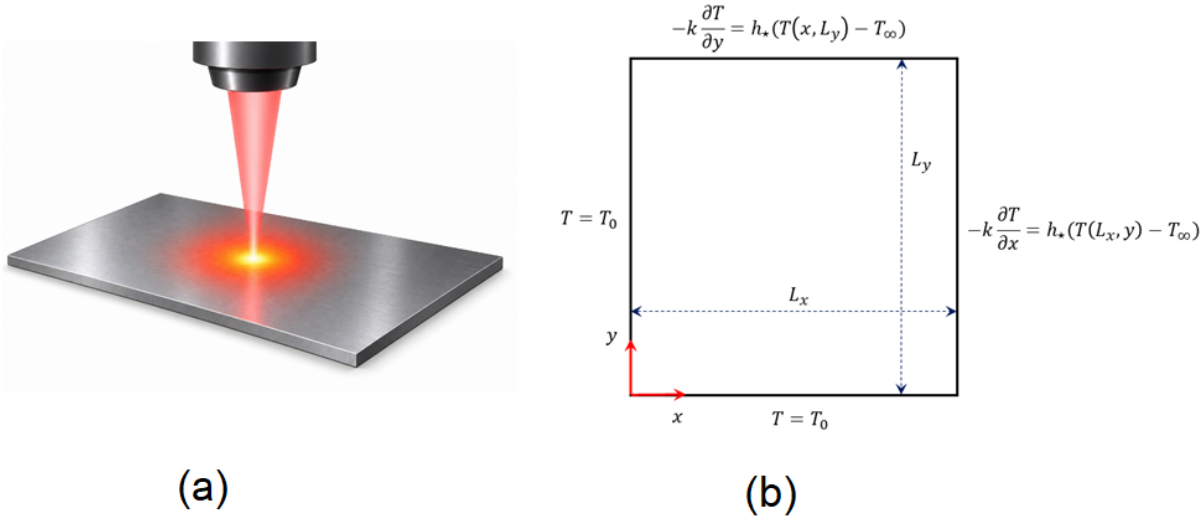


Figure 1: Schematic for CFD Project-II.

the ambient air so that the plate managed to reach steady state. The bottom and the left edges of the plate (as shown in panel (b) in the figure) are being maintained at a constant temperature T_0. The ambient air is at temperature T_∞, and the effective heat transfer coefficient between the plate and the ambient air is h_*. The thermal conductivity of the plate material is k. Because, the plate is sufficiently thin, the heat from the laser may be effectively treated as an internal heat generation term within the 2D plane of the plate. The plate has lateral dimensions of $L_x \times L_y$ (see figure). As such, neglecting any temperature variations along the depth of the plate, show that the energy equation boils down to the following:

$$k\nabla^2 T + q'''(\mathbf{x}) = 0 \tag{P2.1}$$

This is subject to the following boundary conditions:

$$T(x, 0) = T_0, \qquad T(0, y) = T_0 \tag{P2.2a}$$

$$\left(k\frac{\partial T}{\partial y}\right)_{x, y=L_y} + h_*(T(x, L_y) - T_\infty) = 0 \tag{P2.2b}$$

$$\left(k\frac{\partial T}{\partial x}\right)_{x=L_x, y} + h_*(T(L_x, y) - T_\infty) = 0 \tag{P2.2c}$$

The heat generation term, which comes from the heat treatment of the plate, may be modelled as:

$$q'''(\mathbf{x}) = \tilde{S}_0 \exp\left\{-\frac{(x - x_0)^2 + (y - y_0)^2}{d^2}\right\}, \tag{P2.3}$$

where S_0 is the intensity of the laser (or, heat source), $(x_0,\ y_0)$ denotes the position on the plate where the laser/heat source is pointed at, and d is proportional to the width of the heat affected zone, which is also related to the width of the laser/heat source.

Let us non-dimensionalize the system as follows:

$$\phi = \frac{T(\mathbf{x}) - T_\infty}{T_0 - T_\infty}, \quad \bar{x}, \bar{y} = \frac{x, y}{L_x} \tag{P2.4}$$

Show that the system of equations in Eqs. (P2.1) - (P2.3) may be expressed in terms of the dimensionless variables as:

$$\frac{\partial^2 \phi}{\partial \bar{x}^2} + \frac{\partial^2 \phi}{\partial \bar{y}^2} + \bar{q}'''(\bar{\mathbf{x}}) = 0, \quad 0 \le \bar{x} \le 1, \quad 0 \le \bar{y} \le \bar{L}_y \tag{P2.5}$$

where $\bar{L}_y = L_y/L_x$, and

$$\bar{q}'''(\bar{\mathbf{x}}) = S_0 \exp\left\{-\frac{(\bar{x} - \bar{x}_0)^2 + (\bar{y} - \bar{y}_0)^2}{\bar{d}^2}\right\}, \quad S_0 = \frac{\tilde{S}_0 L_x^2}{k}. \tag{P2.6}$$

Further, show that the boundary conditions in Eq. (P2.2) simplify to:

$$\phi(\bar{x}, 0) = \phi(0, \bar{y}) = 1 \tag{P2.7a}$$

$$\left(\frac{\partial \phi}{\partial \bar{y}}\right)_{\bar{x}, \bar{y} = \bar{L}_y} + h\phi(\bar{x}, \bar{L}_y) = 0 \tag{P2.7b}$$

$$\left(\frac{\partial \phi}{\partial \bar{x}}\right)_{1, \bar{y}} + h\phi(1, \bar{y}) = 0, \tag{P2.7c}$$

where $h = h_* L_x/k$. The idea is to solve Eq. (P2.5), subject to Eqs. (P2.6) and (P2.7).

While we are at it, we may as well try to solve a more general equation of the form:

$$\nabla^2 \phi = f_1(x, y)\phi + f_2(x, y), \quad 0 \le x \le L_x, \quad 0 \le y \le L_y, \tag{P2.8}$$

subject to,

$$\phi(x, 0) = \phi(0, y) = \phi_0 \tag{P2.9a}$$

$$\left(\frac{\partial \phi}{\partial y}\right)_{x, y = L_y} + h_2\phi(x, L_y) = 0 \tag{P2.9b}$$

$$\left(\frac{\partial \phi}{\partial x}\right)_{L_x, y} + h_1\phi(L_x, y) = 0. \tag{P2.9c}$$

Any arbitrary forms of the functions $f_1(x, y)$ and $f_2(x, y)$ may be taken as the use's choice. **Write a Finite Difference Based code to solve Eq.** (P2.8) **subject to the boundary conditions** (P2.9)**. As a test case, use your code to solve the heat transfer problem in the plate mentioned earlier, i.e., solve Eq.** (P2.5)**, subject to the boundary conditions** (P2.7)**, and Eq.** (P2.6) **for** $q'''(\mathbf{x})$**.** You are free to choose the values of the various parameters. As output, plot the contours of $\phi(x, y)$.

Also plot ϕ vs x for at various y-locations, and ϕ vs y at various x locations.

Tentative Weightage of the Project: 6% of the entire course.